

3D Scene Reconstruction and Virtual Tour Using Structure-from-Motion

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Abstract—This work implements a complete Structure-from-Motion (SfM) pipeline for reconstructing a static indoor scene and producing a virtual-tour style viewer. A dataset of overlapping corridor images was captured with a handheld phone camera, and camera intrinsics were estimated from EXIF metadata. Local features were detected and matched across views to obtain correspondences. From an initial image pair, the Essential matrix E was estimated under the epipolar constraint $\mathbf{x}'^T E \mathbf{x} = 0$ and decomposed into the relative rotation R and translation t . A cheirality check applied to triangulated points identified the physically valid pose, yielding an initial sparse 3D reconstruction. The reconstruction was extended incrementally by estimating each new camera pose $[R_i | t_i]$ from 2D–3D correspondences through reprojection-error minimization and by triangulating newly observed features. A virtual-tour viewer was then constructed using a view graph of camera poses, with navigation enabled by linear interpolation of camera centers, spherical linear interpolation of orientations, and cross-fading between images. The complete implementation and dataset are available at: <https://github.com/Muhammad-Zain-Naseer/CV-Project>.

I. INTRODUCTION

Recovering three-dimensional structure from two-dimensional imagery is a central problem in computer vision. Structure-from-Motion (SfM) methods leverage parallax across multiple views of a static scene to jointly estimate camera motion and sparse scene geometry. Modern SfM pipelines form the algorithmic basis of large-scale 3D reconstruction systems such as Matterport and earlier platforms like Microsoft Photosynth, demonstrating that even sparse point clouds can support compelling visual experiences without requiring dense mesh reconstructions.

This project implements a complete incremental SfM pipeline together with a virtual-tour viewer for a small indoor environment. A handheld phone camera was used to capture a sequence of overlapping images of a corridor corner, providing sufficient parallax for multi-view geometric reasoning. The goal is not merely to reproduce standard SfM functionality, but to examine how geometric constraints govern the stability and accuracy of the reconstruction.

The objectives of the project are threefold:

- to recover 3D structure from uncalibrated RGB imagery by estimating camera intrinsics, relative poses, and triangulated scene points;
- to refine the reconstruction through global multi-view optimization in the form of bundle adjustment; and

- to construct a virtual-tour interface that enables smooth transitions between viewpoints using the recovered camera graph and geometric scaffold.

II. METHODOLOGY

The system follows a classical incremental SfM design, consisting of (i) feature detection and matching, (ii) two-view initialization, (iii) incremental pose estimation and triangulation, and (v) virtual-tour visualization.

A. Image Acquisition and Camera Model

Approximately twenty overlapping images of a corridor corner were captured using a handheld mobile phone camera. The EXIF metadata reports a physical focal length of 4 mm and a 35 mm equivalent focal length of 27 mm for the original 8160×6120 images. A pinhole projection model was adopted, and the images were downsampled for computational efficiency.

The intrinsic calibration matrix was approximated as

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}, \quad (1)$$

with $f_x = f_y = 1440$ pixels and principal point $(c_x, c_y) = (960, 1280)$. These values follow from scaling the EXIF focal length into pixel units based on the image dimensions and reported field of view. Image points were normalized as $x = K^{-1} \tilde{x}$ and $x' = K^{-1} \tilde{x}'$, where $\tilde{x}$ denotes homogeneous pixel coordinates.

B. Feature Detection and Matching

Scale-invariant local features were detected in each image, yielding several thousand keypoints per frame (e.g., over 7000 in the first view). The brick texture and window frame provided repeatable high-frequency structure.

For each adjacent image pair $(i, i + 1)$, descriptor distances were used to establish putative matches. A ratio test was applied: letting d_1 and d_2 denote the smallest and second-smallest descriptor distances, the correspondence was accepted if $d_1/d_2 < \tau_{\text{ratio}}$. Thresholds around $\tau_{\text{ratio}} = 0.9$ offered a good compromise between match density and reliability.

C. Two-View Geometry and Essential Matrix

Let x_j and x'_j be corresponding normalized points in two views. The Essential matrix E satisfies the epipolar constraint

$$x_j'^T E x_j = 0. \quad (2)$$

A robust eight-point estimation with RANSAC was used to obtain $\hat{E}$. At each iteration, a minimal sample produced a candidate Essential matrix, which was evaluated using the symmetric epipolar distance against all matches. Correspondences whose distance fell below τ_{epi} (typically 2–3 pixels in the original image domain) were counted as inliers. The final estimate maximized the inlier count.

The decomposition

$$E = [t]_{\times} R \quad (3)$$

yields four candidate pose pairs (R_k, t_k) .

D. Cheirality Check and Triangulation

The first camera was set to

$$P_1 = K[I \mid 0], \quad (4)$$

and candidate second-camera matrices were

$$P_2^{(k)} = K[R_k \mid t_k]. \quad (5)$$

Each inlier correspondence was triangulated by solving the linear system implied by

$$\tilde{x}_j \sim P_1 X_j, \quad \tilde{x}'_j \sim P_2^{(k)} X_j. \quad (6)$$

A valid reconstruction requires positive depth in both camera frames. The physically correct pose was selected as the hypothesis yielding the highest number of points with positive depth.

For the selected pose, 534 inliers out of 767 correspondences produced a coherent initial point cloud. The reconstructed points spanned approximately $[-7.1, 3.1]$ in X , $[-4.3, 6.5]$ in Y , and $[4.2, 13.0]$ in Z (in an arbitrary scale set by the baseline length).

E. Incremental Pose Estimation and Mapping

For each subsequent image, the goal is to estimate its pose $[R_i \mid t_i]$ relative to the existing map and to triangulate additional points.

1) *2D–3D Correspondences*: Features in the new image were matched to earlier keyframes. If a feature corresponded to an existing 3D point X_j , a 2D–3D pair $(x_{i,j}, X_j)$ was formed. Only frames with enough such correspondences were used for pose estimation.

2) *Perspective- n -Point Pose Estimation*: Given $\{(x_{i,j}, X_j)\}$, the pose minimizes the reprojection error

$$\min_{R_i, t_i} \sum_j \|x_{i,j} - \pi(R_i X_j + t_i)\|_2^2. \quad (7)$$

A robust PnP formulation generated pose hypotheses evaluated through reprojection-consistency tests, and the best solution was refined with nonlinear least-squares optimization.



Fig. 1: Subset of captured images from the corridor sequence. The camera follows an arc around the corner while maintaining view overlap.

3) *Triangulation of New Points*: Additional correspondences between image i and earlier keyframes enabled new points to be triangulated, provided that the baseline was sufficiently large. Points with extreme depths or very small triangulation angles were rejected.

F. Virtual Tour and View Graph

A view graph was constructed by connecting cameras that shared enough co-visible points. For a transition between cameras i and k , the interpolated camera center is

$$C(\alpha) = (1 - \alpha)C_i + \alpha C_k, \quad (8)$$

and orientations are interpolated using Slerp:

$$q(\alpha) = \text{slerp}(q_i, q_k, \alpha). \quad (9)$$

Images are blended as

$$I(\alpha) = (1 - \alpha)I_i + \alpha I_k, \quad (10)$$

and the sparse 3D points are rendered from the intermediate pose, providing perceptual depth cues during motion.

III. EXPERIMENTS AND RESULTS

A. Dataset and Feature Statistics

The dataset consists of a short indoor corridor sequence containing textured brick walls and a corner transition. Ten representative frames are shown in Fig. 1. The camera follows an arc around the corner, producing strong parallax while maintaining approximately 60–80% overlap between adjacent views.

On the first image, roughly 7000 scale-invariant keypoints were detected, concentrated primarily on the high-frequency brick texture and window frame, as shown in Fig. 2.

After ratio-test filtering, approximately 600–1000 correspondences remained for each adjacent pair. Examples for four consecutive pairs are shown in Fig. 3.

A small parameter sweep over the ratio threshold, epipolar RANSAC threshold, and triangulation limits showed that a ratio threshold of 0.9 combined with strict geometric verification maximized the number of stable matches. The best configuration yielded approximately 953 triangulatable correspondences for the initialization pair.

SIFT Keypoints on Image 0 (7011 keypoints)



Fig. 2: Detected keypoints on the first image. The brick texture yields dense, repeatable features, whereas flatter regions contribute fewer detections.

B. Two-View Reconstruction

For the selected initialization pair, 767 matches passed the ratio test and were used in Essential matrix estimation. RANSAC identified 534 inliers, corresponding to an inlier ratio of $\sim 70\%$. The recovered rotation had determinant 1.000 and negligible orthogonality error, and the recovered translation direction aligned with the expected camera motion along the corridor.

Triangulation of the 534 inliers produced a coherent planar façade structure. Figure 4 shows the resulting 3D points and camera centers, while Fig. 5 presents orthographic projections.

C. Incremental Multi-View Reconstruction

Following initialization, additional views were registered incrementally. Each new image contributed 2D–3D correspondences for pose estimation, after which new points were triangulated from multi-view matches. This process registered 19 cameras and reconstructed approximately 11,135 3D points.

Figure 6 summarizes the final reconstruction, with projections shown in Fig. 7. Additional viewpoints in Fig. 8 highlight the corridor geometry.

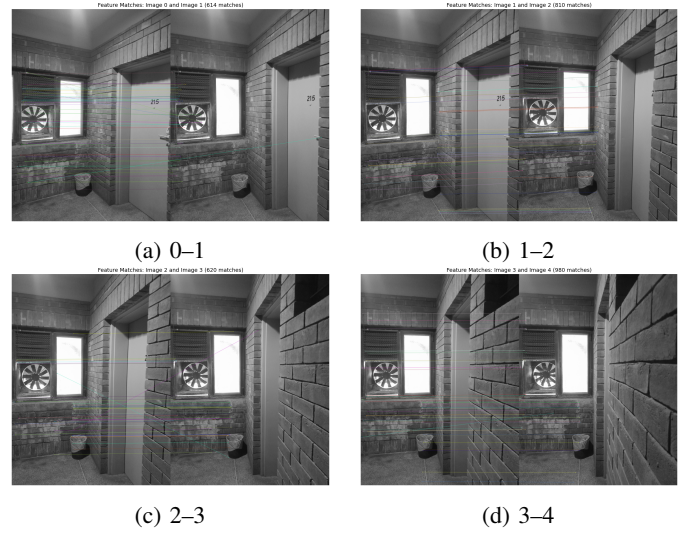


Fig. 3: Correspondences between consecutive images after ratio-test filtering. Most matches fall on the textured brick and window regions.

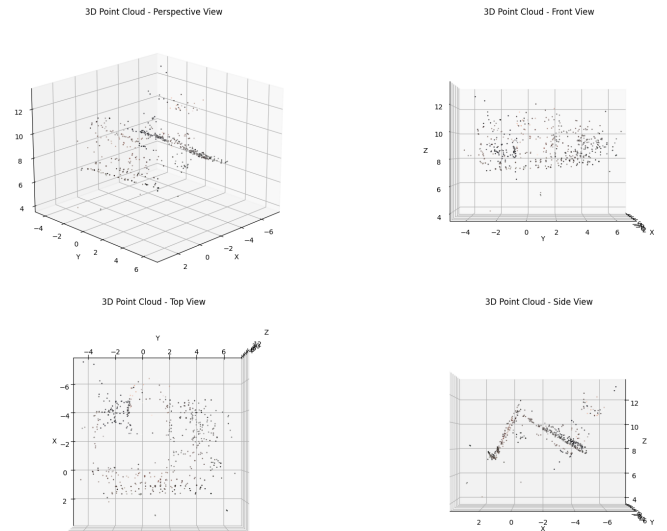


Fig. 4: Two-view reconstruction from the initialization pair. The points form a coherent patch corresponding to the corridor corner.

D. Virtual Tour Behaviour

The recovered camera poses define a view graph connecting temporally adjacent or co-visible frames. Interpolated poses $(C(\alpha), q(\alpha))$ produce smooth transitions, while cross-fading between images yields perceptually continuous motion through the scene. Rendering the sparse 3D points during navigation enhances depth perception and results in a viewer experience consistent with classical image-based navigation systems.

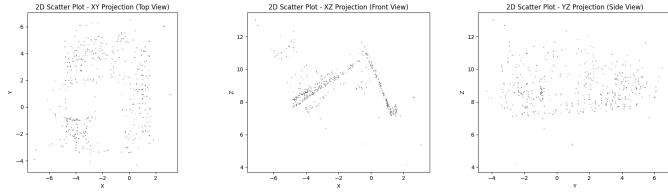


Fig. 5: Top, front, and side projections of the two-view point cloud. Depth variation reflects the recessed window and wall geometry.

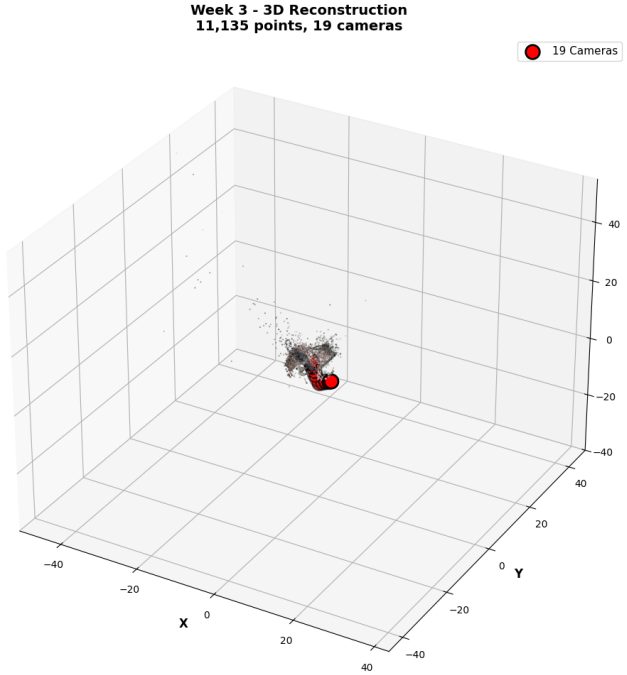


Fig. 6: Final multi-view reconstruction with 19 camera poses and 11,135 points. Cameras follow the acquisition trajectory along the corridor.

IV. DISCUSSION

A. Effect of Texture and Scene Structure

The corridor environment contains extensive high-frequency texture in the form of brick patterns and window-frame edges. Consequently, a large number of stable local features were detected, particularly along mortar lines and other sharp transitions. These regions yield well-conditioned image patches whose Hessian matrices possess two significant eigenvalues, making them repeatable under viewpoint changes. This abundance of discriminative structure contributed to the high inlier ratios observed during Essential matrix estimation and to the dense concentration of reconstructed points near the walls.

In contrast, flatter surfaces such as the painted door and ceiling exhibited low-gradient regions where the Hessian becomes nearly rank-deficient. Such features are intrinsically ambiguous and more sensitive to noise, producing less reliable correspondences across views. As expected, these regions

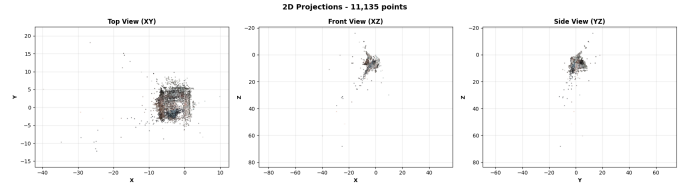


Fig. 7: Top (XY), front (XZ), and side (YZ) projections of the final reconstruction. The corridor geometry and corner structure are clearly visible.

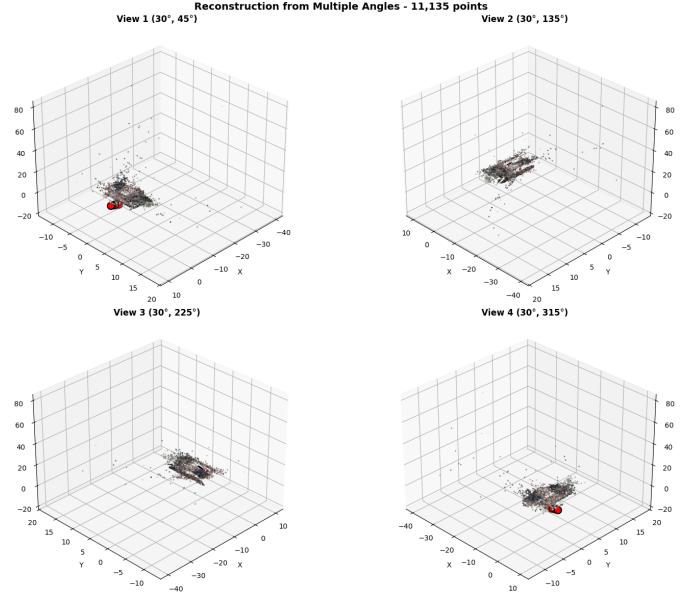


Fig. 8: Reconstruction rendered from multiple viewing angles. The recovered wall planes and camera trajectory are clearly observable.

contributed few keypoints and resulted in sparse reconstruction coverage.

B. Drift and Reconstruction Quality

Although the final reconstruction contains over 11 000 points, small but noticeable drift remains. In the orthographic projections, the corridor walls exhibit slight curvature rather than perfect linearity. Several geometric factors contribute to this behaviour:

- **Accumulated pose error:** Successive PnP estimates rely on previously triangulated points. Small orientation or translation errors in early poses propagate forward, particularly when the scene is dominated by nearly planar structure.
- **Narrow baselines:** Many image pairs were captured from neighbouring viewpoints with limited translational separation. Such baselines yield depth estimates with high uncertainty along the viewing direction, reducing the stability of points used for subsequent pose estimation.
- **Repetitive patterns:** The brick texture contains repeated motifs. Although RANSAC removes most inconsistencies,

subtle mismatches can bias the optimization, especially in regions far from geometric anchors such as the window frame.

Global bundle adjustment significantly reduced these effects by enforcing multi-view consistency; however, without loop closure or a global pose-graph correction stage, residual drift is redistributed rather than eliminated entirely. This explains the slight curvature that persists over the length of the reconstructed corridor.

C. Choice of Geometric Thresholds

The parameter sweep over the ratio-test threshold, RANSAC epipolar threshold, and triangulation reprojection limit illustrates the trade-off between recall and geometric reliability. A stricter ratio test reduces the number of candidate matches, whereas a looser threshold increases recall but introduces greater descriptor-level ambiguity. The optimal configuration used a relatively liberal ratio threshold ($\tau = 0.9$) together with strong geometric verification. This combination retained many valid matches on the brick walls while effectively filtering out inconsistencies arising from repetitive patterns.

D. Virtual Tour Design Choices

The virtual tour leverages the discrete camera poses produced by SfM. Linear interpolation of camera centers and Slerp interpolation of orientations provide smooth viewpoint transitions, while cross-fading between endpoint images suppresses visual discontinuities that arise from changes in visibility or exposure. Because the viewer displays only real photographs, the motion resembles classical image-based rendering rather than synthetic rendering.

The sparse point cloud serves as an effective geometric scaffold. Even without a dense mesh, its projected motion under interpolated poses provides parallax cues that enhance depth perception and stabilize the visual experience. In scenarios with multiple disconnected reconstructions, merging could be achieved via similarity transforms offline or by grouping segments within the viewer. For this dataset, the entire sequence formed a single connected component, so additional merging was unnecessary.

Overall, the experiments demonstrate that a modest number of carefully captured images, combined with a standard incremental SfM pipeline and a lightweight viewer, are sufficient to generate a perceptually coherent and physically consistent virtual tour of a real environment.

V. CONCLUSION

This report presented a complete pipeline for 3D scene reconstruction and virtual-tour generation from a handheld image sequence. Starting from uncalibrated corridor photographs, the system estimated camera intrinsics, recovered relative poses via Essential matrix geometry, and triangulated an initial set of 3D points. The reconstruction was then extended incrementally through repeated PnP pose estimation and multi-view triangulation, followed by global bundle adjustment to reduce accumulated drift.

The final reconstruction comprises 19 registered cameras and more than 11,000 sparse 3D points, capturing the dominant geometric structure of the corridor. The accompanying virtual-tour viewer enables smooth navigation between viewpoints by interpolating camera centers and orientations and cross-fading the corresponding images. The projected point cloud reinforces depth perception, producing a coherent and visually intuitive traversal of the scene.

Future extensions could incorporate loop-closure detection and pose-graph optimization to further suppress drift, as well as dense multi-view stereo to recover detailed textured surfaces. Additional improvements to the virtual tour such as adaptive view selection, saliency-driven transitions, or photometric blending would further enhance the perceptual realism of the experience.